VACC 201 Tinkering and Mentoring

Name of the Product

TriSense-Heal Patch

A multi-biomarker responsive smart wound dressing for early infection detection

Early Biochemical Warning Before Visible Infection

Only plant/bio-sourced biodegradable materials used

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Objectives

Primary Objectives

To develop a **biodegradable bandage system** using agar-based hydrogel and natural materials
To integrate **multi-parameter sensing**, including:

- pH changes (infection indicator)
- Protease activity (tissue degradation marker)
- Oxidative stress (healing/inflammation marker)
- Moisture level (wound environment)
- Temperature variation (inflammation/infection indicator)

Secondary Objectives

To utilize **natural bioactive compounds**:

- Anthocyanin (pH sensing)
- Bromelain (proteolytic activity & wound cleaning)
- Curcumin (antimicrobial & antioxidant)

To design a **hybrid architecture**:

- Functional hydrogel layer (healing)
- Surface sensor zones (visual detection)

To ensure:

- **Cost-effectiveness**
- **Ease of use (visual color-based detection)**
- **Scalability for mass production**

Applied Objective

To create a **prototype capable of early infection detection**, reducing dependence on delayed visual diagnosis and enabling timely medical intervention.

Problem Statement

Conventional bandages are passive, non-biodegradable, and lack real-time monitoring, leading to delayed infection detection. This creates a need for a biodegradable smart bandage capable of monitoring multiple wound parameters simultaneously.

CONCEPT OVERVIEW (GRAPHICAL REPRESENTATION)

ARCHITECHTURE

Low-Cost, Natural Biomaterial Sourcing

Replacing expensive synthetic lab chemicals with highly effective, biodegradable natural equivalents.

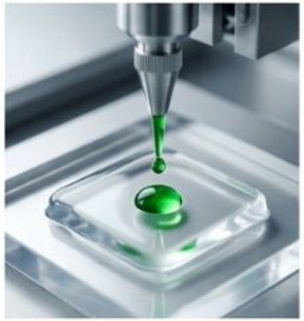
Bioactive Components	Structural Base	Sensor Indicators
 **Turmeric (Curcumin)** - Antioxidant properties and ROS redox response.	 **Gelatin / Alginate (100–200 g)** - Hydrogel base for sensing zones (Approx. Cost: 800 INR).	 **Red Cabbage Extract (Anthocyanin)** - Natural pH-sensitive colorimetric indicator.
 **Pineapple Extract (Bromelain)** - Natural enzyme integration for protease interactions.	 **Sterile Cellulose Gauze** - Soft, absorbent contact layer.	 **Resazurin / Bromothymol Blue** - Supplemental oxidative and pH validation dyes.

Laboratory Prototype Development

Translating accessible raw biomaterials into a precisely engineered, functional diagnostic dressing.

1. Extraction: Beakers containing vibrant extracted natural anthocyanin dyes.

2. Base Formation: Liquid hydrogel poured into precise acrylic molds.



3. Sensor Integration: Micro-dispensing the circular sensor zones.

4. Final Assembly: The fully assembled prototype patch in its dry state.

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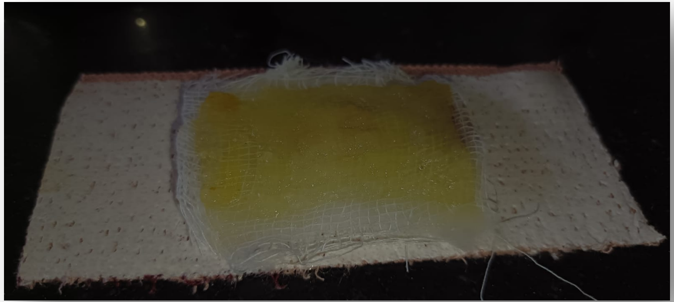
PROTOTYPE DEVELOPMENT

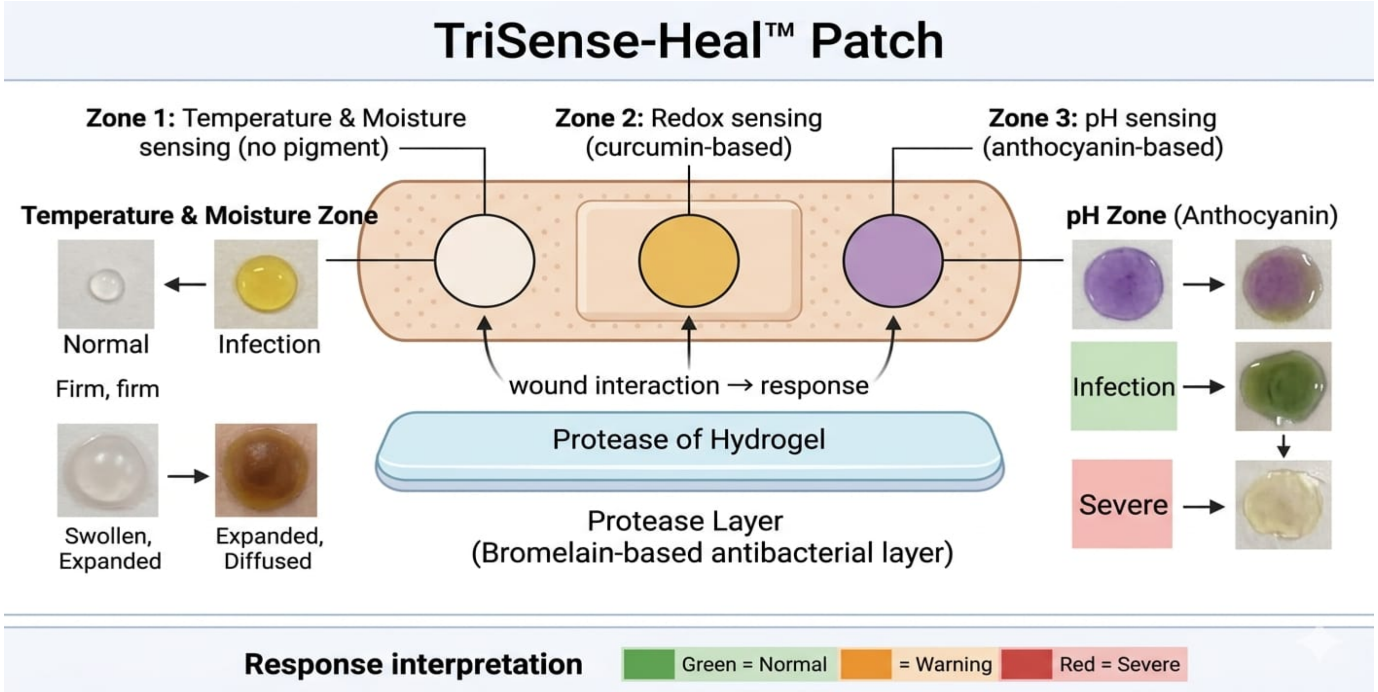
PROTOTYPE



PROTEASE LAYER

FOIL PACKAGING

ILLUSTRATION ON PACKAGING



OBSERVATION TABLE OF VISIBLE RESPONSE

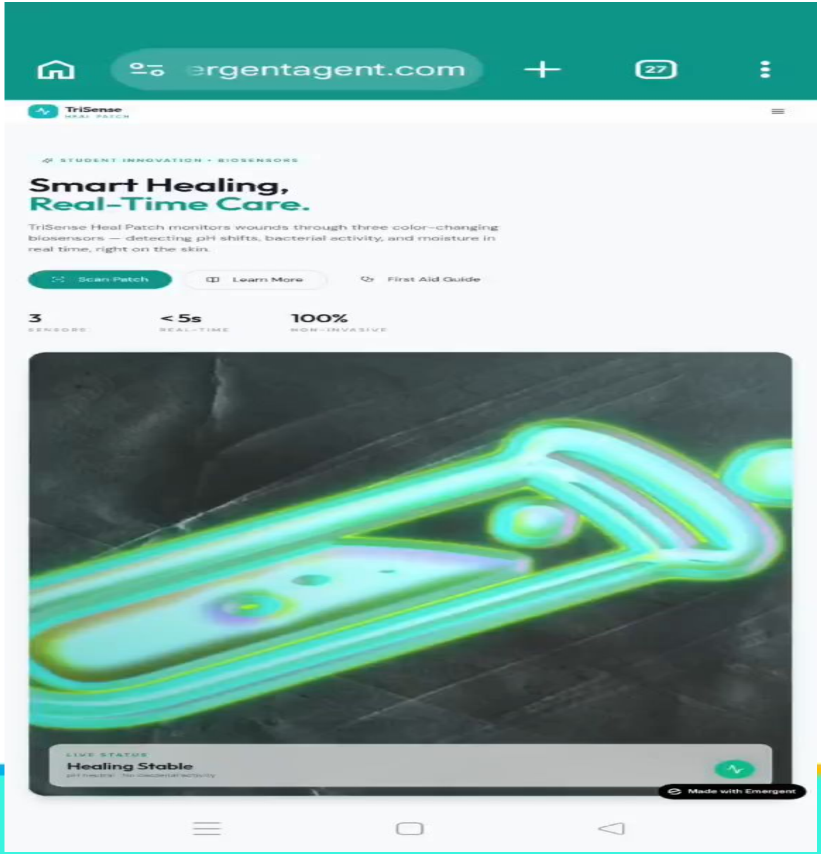
	Sensor	Trigger	Before	After
1	pH Variation	Wound shifts to alkaline	Purple	Green/Yellow
2	Protease Activity	Bacterial enzymes degrade gelatin	Contained Dye	Matrix Breakdown & Spread
3	Oxidative Stress	Elevated ROS from inflammation	Baseline	Intensity Change via Redox
4	Moisture	Excessive exudate buildup	Dry Strip	Physical Swelling
5	Temperature	Localized inflammatory heat	Neutral	Thermochromic Shift

CHALLENGES AND IMPROVEMENTS

Challenge	Improvement
Hydrated Gel Shelf-Life Limitations	**Dry-State Activation** The patch remains inert during storage and activates only upon contact with wound exudate.
Natural Biomolecule Degradation	**Micro-Encapsulation** Shielding active ingredients like curcumin to prevent premature fading and ensure long-term molecular stability.
Environmental Contamination Risk	**Advanced Packaging** Utilizing medical-grade vacuum sealing to ensure complete sterility and baseline stability until application.

WOUND CARE – SMART WOUND CARE

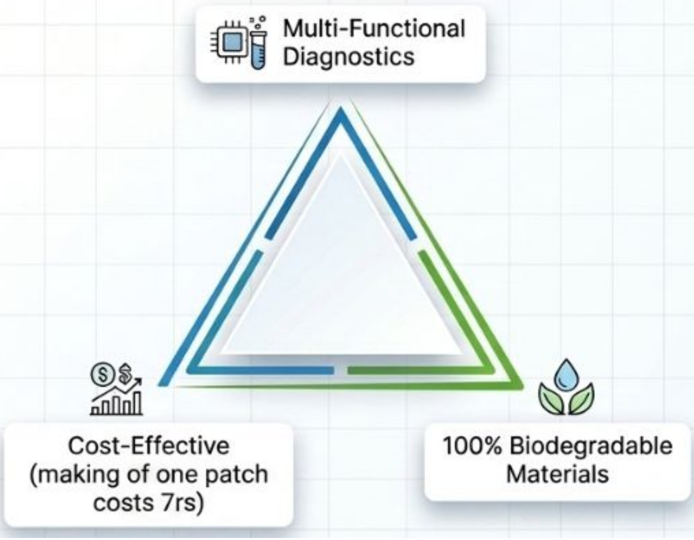
SCAN TO TRACK YOUR WOUND AND PREDICT INFECTION RATE



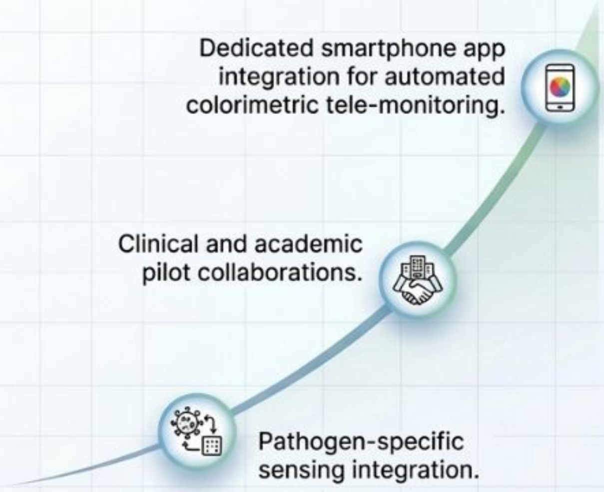
The Future of Diagnostic Wound Care

A scalable, single-use consumable that shifts wound care from passive observation to proactive medical intervention.

The Value Triangle



Development Roadmap



…THANK YOU